Glycaemic responses to glucose and rice in people of Chinese and European ethnicity.

AIMS: Diabetes rates are especially high in China. Risk of Type 2 diabetes increases with high intakes of white rice, a staple food of Chinese people. Ethnic differences in postprandial glycaemia have been reported. We compared glycaemic responses to glucose and five rice varieties in people of European and Chinese ethnicity and examined possible determinants of ethnic differences in postprandial glycaemia. METHODS: Self-identified Chinese (n = 32) and European (n = 31) healthy volunteers attended on eight occasions for studies following ingestion of glucose and jasmine, basmati, brown, Doongara(®) and parboiled rice. In addition to measuring glycaemic response, we investigated physical activity levels, extent of chewing of rice and salivary α-amylase activity to determine whether these measures explained any differences in postprandial glycaemia. RESULTS: Glycaemic response, measured by incremental area under the glucose curve, was over 60% greater for the five rice varieties ($P < 0.001$) and 39% greater for glucose ($P < 0.004$) amongst Chinese compared with Europeans. The calculated glycaemic index was approximately 20% greater for rice varieties other than basmati ($P = 0.01$ to 0.05). Ethnicity [adjusted risk ratio 1.4 (1.2-1.8) $P < 0.001$] and rice variety were the only important determinants of incremental area under the glucose curve. CONCLUSIONS: Glycaemic responses following ingestion of glucose and several rice varieties are appreciably greater in Chinese compared with Europeans, suggesting the need to review recommendations regarding dietary carbohydrate amongst rice-eating populations at high risk of diabetes.